ABFormer: A Transformer-based Model to Enhance Antibody-Drug Conjugates Activity Prediction through Contextualized Antibody-Antigen Embedding

Rushi Katabathuni[1, 2], Viren Loka[1, 2], Sanjana Gogte[2], Vani Kondaparthi[*2, 3]

[1]Department of Computer Science and Engineering, Keshav Memorial College of Engineering, Ibrahimpatnam, Hyderabad, Telangana, India-501510,

[2] Drugparadigm Research Lab, Uppal, Hyderabad, Telangana, India-500088,

[3]Department of Chemistry, Keshav Memorial Engineering College, Uppal, Hyderabad, Telangana, India – 500088.

*Corresponding Author

Email: vani@drugparadigm.in

Abstract:

The design of effective Antibody-Drug Conjugates (ADCs) remains a challenge due to intertwined interactions amongst various components. Best-in-class computational models like ADCNet typically predict ADC activity, treating antibody, antigen, and drug moieties as independent features. This overlooks, to a significant extent, dynamic antibody-antigen binding interface interactions that play central roles in determining ADC specificity and internalization. Towards this, we propose ABFormer, a new transformer-based architecture that addresses this omission by incorporating contextual antibody-antigen binding features, using a re-purposed AntiBinder model as input for the ADC activity prediction workflow. This eliminates naive, independent component embeddings with a high-rank, interaction-perceptive feature set capable of encoding high-quality molecule recognition dynamics. On the publicly available ADCdb dataset, ABFormer demonstrated significantly improved performance over the control ADCNet, achieving an accuracy of 85.61% compared to 72.73%, a balanced accuracy of 84.25% compared to 70.88%, and a PRAUC of 92.88%. In an ablation experiment, the role of the contextual binding unit was confirmed, as its elimination resulted in a notable MCC reduction from 0.6854 to 0.2251. ABFormer also attained 100% accuracy in an independent external test set comprising 21 novel ADCs, hence outperforming ADCNet (90.48%). Overall, these results highlight the need to model interdependent ADC component dynamics for accurate activity forecasting, thereby establishing a viable computational platform that outperforms classical feature-driven ADC architectures.

Keywords: Antibody Drug Conjugates (ADCs), Transformer Architecture, ADCNet, Deep Learning, ADC Activity Prediction, Antibody-Antigen Binding, Computational Modeling, Contextual Features, Drug Design, ADCdb.

1. Introduction:

The extremely complex process of discovery of drugs involves a sensitive equilibrium among pharmacokinetics, therapeutic performance, as well as issues of safety. Despite the optimizing potential that is provided through calculations, high attrition levels are experienced from time to time with consequent instances commonly resulting from unexpected toxicity as well as target failure. [1]. Antibody-Drug Conjugates (ADCs) proved to be highly effective as a therapy platform to overcome these challenges [2]. By linking a highly potent cytotoxic

payload with a tumor-specific monoclonal antibody using a stable linker molecule, ADCs deliver chemotherapy in a targeted fashion directly to the cancer cells.

Background on Antibody-Drug Conjugates (ADCs)

Antibody-Drug Conjugates (ADCs) are site-specific biopharmaceuticals that are developed to deliver highly potent cytotoxins precisely to tumor cells. At times also referred to in publications as a "biological missile," therapy utilizes monoclonally mediated specificity to increase anti-tumor efficacy while simultaneously decreasing systemic toxicity, thus holding a significant advantage over traditional chemotherapy [1]. Effective ADC design involves subtle optimization of these three essential components: a monoclonal antibody, chemical linker, and cytotoxic payload. [2].

Key Components and Mechanism

1. **The Monoclonal Antibody (mAb):** This part serves as the targeting vehicle that is carefully fabricated to bind with good specificity to a specific Tumor-Associated Antigen (TAA) that is overexpressed on cancer cells while showing weak expression on healthy cells. This selective binding is the first and foremost important step involved in targeted drug delivery [3].

2. **The Linker:** The chemical bridge connects payload covalently with antibody. The design of this bridge is essential for conferring ADC with stability under circulation as well as with releasing payload on a controlled schedule only when it encounters the target cell[4]. Linkers can be broadly divided into two classes: cleavable linkers, which are cleaved off under certain defining circumstances within the tumor microenvironments (such as levels of acidity or certain proteases), and non-cleavable linkers, which release the payload only upon cleavage of the antibody within the cell.

3. **The Cytotoxic Payload:** The cytotoxic payload is a high potency small molecule that kills cells. It must possess potency in the picomolar to nanomolar range, necessitated by the low intracellular concentration achieved upon delivery. [5]. Payloads disrupt important cell processes, for example, microtubule dynamics (e.g., auristatin) or by causing DNA damage (e.g., calicheamicin) [6, 7]. Payload also must possess appropriate chemical characteristics for stable linker conjugation as well as low systemic toxicity if it is released unconjugated.

ADC effectiveness as well as safety are also determined by Drug-to-Antibody Ratio (DAR). [8].

Mechanism of Action of ADCs

The overall mechanism by which Antibody-Drug Conjugates (ADCs) work permits targeted cytotoxicity via a series of systematic processes [9]. Initially, the ADC remains intact in circulation until its monoclonal antibody moiety preferentially binds to its corresponding antigen on the cell membrane of a cancer cell. Binding initiates receptor-mediated endocytosis, which results in internalization of the ADC-antigen complex into endosomes. [10]. Then, they merge with lysosomes, which are acidic, enzyme-dense compartments. It is here that cleivable or non-cleivable payloads are released: cleivable linkers are broken down by acidic or lysosomal enzyme environments, while for non-cleivable linkers, the antibody itself gets broken down, releasing the payload still attached to a linker-amino acid molecule. [10]. Ultimately, the released cytotoxic payload or active metabolite gains access to the nucleus and/or cytoplasm to exert its pharmacology. Such event normally entails disruption of key cellular processes such as microtubule polymerization or DNA synthesis, culminating in irreversible cell injury, cell cycle arrest, and apoptosis. [11].

The increasing complexity of therapy design has necessitated the use of deep learning methods in drug discovery. Recent advances involve the de novo design of antigen-specific antibodies with generative models such as diffusion models [12, 13] and large language models [14], the application of protein language models for the swift evolvability of antibodies [15], and the assembly of high-end neural networks for the predictive modeling of drug-target and antibody-antigen affinities [16-19]. Even though such tools show remarkable efficiency in maximizing discrete aspects or approximating pairwise interactions, a main lack remains in modeling the entire system of an antibody-drug conjugate (ADC), where the interaction among all constituents ultimately defines the terminal biological activity [20].

Existing computational models for the prediction of ADC activity have generally treated the antibody, linker, and drug individually as sets of features [21]. Such are typically processed independently and then concatenated, thereby neglecting the rich, biologically meaningful interactions between the antibody and the target antigen [22]. Since these binding events are key determinants of an ADC's specificity and internalization, failing to model them explicitly limits both predictive potential and biological interpretability. In this sense, such a limitation

strengthens the requirement for a model that adequately portrays the interdependent and dynamic interactions between constituents of an ADC and its target. This limitation directly bears on our main research hypothesis: that ADC activity's predictive accuracy may be significantly improved through a contextual embedding of the antibody-antigen binding interface as opposed to treating them as independent features.

To examine this hypothesis, we suggest ABFormer, a transformer model that predicts antibody-drug conjugates' (ADCs) activity based on contextual information that encapsulates antibody-antigen interface from the AntiBinder model [17]. It adopts a knowledge transfer approach to address the limited availability of ADC data, enabling precise predictions with high accuracy and enhancing generalizability [23]. We employed a comprehensive battery of evaluation metrics, including ACC, AUC-ROC, BA, and PRAUC, further establishing ABFormer's outperformance. We also corroborated this on an independent external collection of 21 recently reported ADCs. In this collection, ABFormer scored 100% accuracy, which outperformed baseline ADCNet's 90.48%. The code is openly available to aid rational ADC design.

2. Materials and Methods

2.1. ADCNet

2.1.1. Dataset

A pre-processed dataset was used that was provided in the ADCNet public repository [21](https://github.com/idrugLab/ADCNet/tree/main). This dataset, which originates from ADCdb [24], was subjected to a data preparation and cleaning pipeline by the ADCNet authors to render it of high quality and consistent.

The analysis began with a stringent filtering phase where only those antibody-drug conjugates (ADCs) with full structural detail remained. This was to guarantee that every entry included four essential data points: the light and heavy chain sequences of the antibody, the antigen sequence, and the isomeric Simplified Molecular Input Line Entry System (SMILES) of the linker as well as payload. All ADCs with partial information were eliminated. The raw data was then cleaned to remove duplicate records and normalize the records.

Critical to the overall procedure was normalization of the bioactivity data. All heterogeneous units for bioactivity, including IC50, EC50, and GI50 values (such as μM, pM, etc.), were uniformly converted into nanomolar (nM). For ADCs with multiple biological activity data points, invalid values marked with ">" or "<" were removed, and the minimum recorded value was selected as the final definitive bioactivity measure. Subsequently, a systematic labeling strategy was applied. ADCs that were already in the market, clinical phase I, II, or III were automatically labeled as "positive." For ADCs still in the investigational stage, their classification was determined by their bioactivity: they were labeled "positive" if their IC50, EC50, or GI50 value was less than or equal to 100 nM, and "negative" otherwise [25].

2.1.2 Feature Representation and Model Architecture

The antigen, antibody, linker, payload, and drug-to-antibody ratio (DAR) are the five essential elements of antibody-drug conjugates that are integrated into the conventional model, a deep learning framework. To extract rich embeddings from these components, it uses two cutting-edge, pre-trained BERT-based models [26]. ESM-2 [27] is a Transformer-based protein language model that creates 1280-dimensional feature vectors from amino acid sequences for protein-based components, including the antigen and the heavy and light chains of the antibody. The corresponding symbols for these vectors are t_{heavy}, t_{light}, and $t_{antigen}$.

For small-molecule components, the recent variant employs FG-BERT [29], as depicted in Figure 1, to process isoSMILES strings and perform functional-group-level analysis of molecular graphs. This produces 256-dimensional embeddings for the linker (x_1^A) and the payload (x_2^A). The isoSMILES are first pre-processed to generate molecular graphs and adjoin matrices.

The final step involves concatenating the features in the order: x_1^A, x_2^A , t_{heavy}, t_{light}, $t_{antigen}$ and t_{DAR}, where t_{DAR} represents the normalized and standardized DAR values. The formula is as follows:

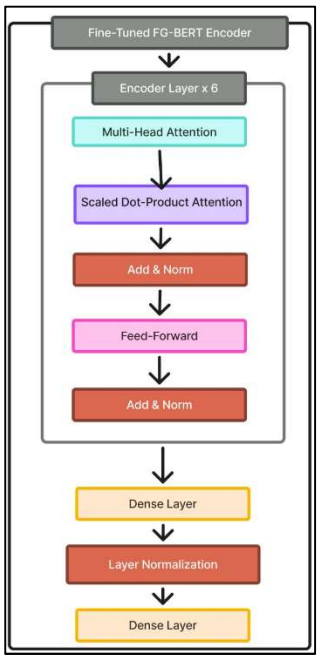
$$x = concat(\mathrm{x}_1^{\mathrm{A}},\ \mathrm{x}_2^{\mathrm{A}}\ ,\ t_{heavy},\ \mathrm{t}_{\mathrm{light}}\ , \mathrm{t}_{\mathrm{antigen}}, t_{DAR}), \qquad (1)$$

This composite feature vector is passed through a Multilayer Perceptron (MLP) with LeakyReLU activation functions [28]. The output logits are then normalized with the output of a sigmoid unit so that we obtain binary classification as corresponding to estimated ADC activity [29].

The MLP formula is stated as follows:

$$\hat{y} = Wx + b, \tag{2}$$

Fig. 1. Architecture Overview of FG-BERT Encoder.



Architecture of Encoder of FG-BERT is presented in Figure 1, which begins with token and position embeddings, then followed appropriately by a stack of encoder layers. The encoder layers are themselves comprised of Multi-Head Attention (MHA) with Scaled Dot-Product Attention, then followed by Add & Layer Normalization, Feed-Forward Network (FFN), then another instance of Add & Layer Normalization being applied last. When complete with encoding through the stack of encoders, dense layers with end-layer normalization are employed to deliver the output of the representations.

2.1.3. Porting FG-BERT from TensorFlow to PyTorch

Noticing the excellence of FG-BERT[30] in learning strong features, we integrated it into our design. This necessitated a reimplementing of the original ADCNet model's TensorFlow-implemented FG-BERT module in PyTorch. We guaranteed architectural accuracy and transformed the original TensorFlow checkpoints to a PyTorch-compatible state dictionary for a straightforward and precise transfer of learned weights.

In order for scalable training of models on multi-GPU architectures, we extended the framework by removing explicit structural encodings, i.e., adjacency matrices, thus allowing for efficient training with PyTorch's Distributed Data Parallel framework[31]. This was a necessity to overcome data collation bottlenecks that arise as a result of non-uniformly sized

graph structures. Pre-trained weights had structural information, but our methodology benefits from the Transformer's in-built self-attention [32, 33] such that it learns novel molecular interdependencies, therefore allowing for efficient parallel training with retention of predictive ability of the model.

2.2. AntiBinder

2.2.1. About the AntiBinder Model

AntiBinder is a deep-learning model designed to predict whether a given antibody heavy-chain sequence will bind to a specific antigen [34, 35]. It integrates both sequence- and structure-based information by first generating high-dimensional embeddings via ESM to capture sequence context and 3D features via IgFold for encoding geometric and contact-based cues [36].

2.2.2. Dataset:

The MET dataset, selected from four existing antibody-antigen interaction datasets for its suitability for cancer therapeutic applications, served as the training dataset for the AntiBinder component. Emibetuzumab, a clinical-stage antibody targeting the MET receptor linked to tumor development and metastasis, has its affinity and specificity maximized using Makowski et al.'s curated MET dataset [37] [38].

A balanced picture of antibody-antigen binding and non-binding interactions is provided by the 4,000 pair sequences in the MET dataset, which include 1,516 positive interacting pairs and 2,484 negative interacting pairs. Considering that this dataset is extremely important to cancer therapies and the ADC application domain, it was chosen over the other available datasets (the COVID-19, HIV, and BioMap datasets). The binding patterns derived from this set are particularly pertinent to ADC design scenarios, as the MET receptor is a therapeutically validated cancer target [39].

The dataset's preparation followed accepted practices for simulating antibody-antigen interactions. Complete heavy chain antibody sequences with the corresponding MET antigen sequence and binary labels that represent the binding interaction make up each pair of sequences. The high-quality curation of this dataset provides stable training signals for the AntiBinder model to acquire informative representations of antibody-antigen binding.

2.2.3. Anti-Binder Pre-processing:

AntiBinder takes antibody heavy-chain and antigen sequences and transforms them into numerical embeddings that capture both sequence and structural cues. The pre-processing starts with splitting antibody heavy chains into regions (FR and CDR) by using the Anarci module [40, 41]. The sequence embeddings of concatenated antibody regions and antigens are taken from ESM-2. Then, antibody & antigen structures are generated with the IgFold model, and then the sequence & structure features are combined.

2.2.4. Overview of Antibinder Architecture:

The high-level schematic of the Antibinder architecture is depicted in Figure. The model begins by passing the antibody and antigen sequences through their respective embedding modules AbSeqEmb and AgSeqEmb to create contextual numerical representations. These embeddings are then fed into a stack of N Bidirectional Attention Blocks (BidAttBlocks) that observe sophisticated inter-molecular interactions [42]. The final representations are fed into a Classifier module, which outputs predictions like binding affinity or classification scores [43].

AbSeqEmb Module: The AbSeqEmb module deals with antibody sequences through two parallel channels:

Sequence-based pathway: This pathway uses Amino Acid (AA) Indexing and Region Indexing to obtain sequence-level and region-specific features (e.g., CDRs), which are then padded or truncated and passed through a trainable embedding layer.

Structure-based pathway: This pathway uses IgFold, a structure-prediction software package, to retrieve 3D structural information from the sequence. Structural features obtained from the program are sliced and linearly projected. The outputs of the two pathways are element-wise summed and then input to another linear transformation to generate the final antibody embedding [44].

AgSeqEmb Module:

The AgSeqEmb module shows antigen processing. Similar to the antibody pathway, it also starts with AA Indexing and padding/truncating. Importantly, it includes ESM-2, a cutting-edge protein language model, to learn context-aware sequence embeddings. The embedding

and ESM-2 pathways are fused element-wise and processed through a linear layer to yield the antigen embedding [45].

Bidirectional Attention Block (BidAttBlock):

The bidirectional interaction modeling happens in the BidAttBlock, explained in Figure 1(d). Every block takes the antibody and antigen embeddings and performs bidirectional cross-attention [46]:

Antibody-to-antigen attention: The antibody embedding is utilized as the Query (Q). At the same time, the antigen embedding supplies the Key (K) and Value (V) such that the model can attend to antigen areas of significance [47].

Antigen-to-antibody attention: In this instance, the model can observe complementary interactions from the antigen's perspective because the antigen embedding serves as the Query, and the antibody embedding acts as the Key and Value [48].

All attention outputs are passed through Layer Normalization [49] and a Linear layer, then summed. This mutual attention mechanism is vital for learning dependent features among molecules [50].

Classifier Module:

The final antibody-antigen representation, as shown in Figure 2, produced by the final BidAttBlock, is fed into the Classifier module. The module is composed of several Multi-Layer Perceptrons (MLPs) [51]. A gating or modulation scheme, represented as Mod: α•, is used to reweight feature importance dynamically. The embedding is fed through a final MLP layer after being flattened to produce predictions, such as binding probabilities or classification results [52].

This modularity makes it easy to integrate both sequence-related as well as structure-related representations seamlessly, thus making Antibinder a powerful and versatile framework for anticipating antibody-antigen interactions [53].

Fig. 2: Complete Workflow of BiDirectional Cross-Attention Component.

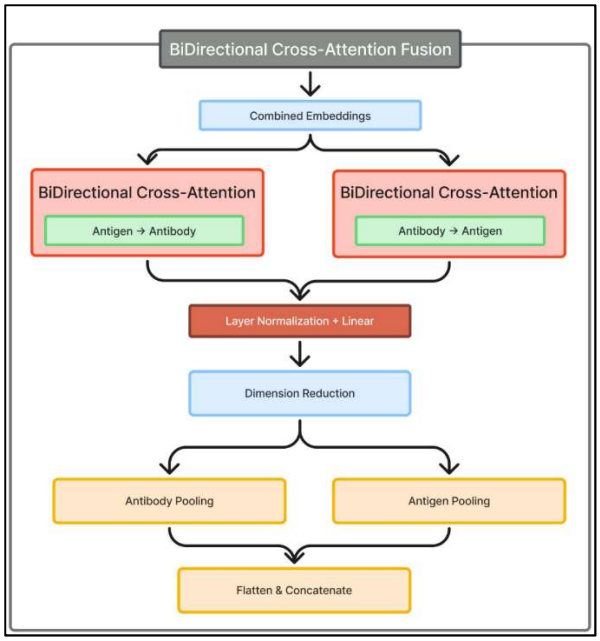


Figure 2 depicts the Bi-Directional Cross-Attention Fusion framework, where antigens' and antibodies' embeddings interact through double cross-attention layers. The integrated features then pass through normalization, dimension reduction, pooling, and concatenation for aiding completion of down-stream prediction tasks.

2.2.5. Model Reconfiguration for Contextual Embedding Extraction

To capture on the molecular nature of antibody-antigen interaction for antibody-drug conjugate (ADC) activity prediction, we changed the architecture of AntiBinder. Initially created for binding result prediction, our work needed a more comprehensive description of the interaction environment. We therefore, after training, eliminated the final scoring layer. This modification enables the network to output interaction-aware, high-dimensional antibody-antigen interaction embeddings that encode applicable molecular recognition information directly for ADC activity prediction [47, 54].

2.3. ABFormer

2.3.1. ABFormer Architecture:

Our multi-channel feature fusion deep learning model, ABFormer combines contextual antibody-antigen binding representations with the original ADCNet architecture, as shown in Figure 3. The main components of the framework are an updated ADCNet for processing ADC-specific features and AntiBinder for simulating antibody-antigen interactions. The architecture employs several pre-trained models to extract detailed features. IgFold is used for structural

modeling of antibody sequences to provide 3D coordinates for AntiBinder analyses, while ESM-2 is used to represent antigen protein sequences. Through transfer learning from pre-training on approximately 1.45 million unlabelled molecules, employing masking-recovery of functional groups, FG-BERT extracts features of small-molecule linkers and payloads.

ABFormer's main innovation is the use of contextual antibody-antigen binding representations, rather than separate antibody heavy, light chain and antigen processing. Specifically, rather than the independent feature vector t_{heavy} used in the baseline ADCNet, we create a combined binding representation $t_{AntiBinder}$ by inputting the antibody heavy chain sequence and antigen sequence separately into AntiBinder, replacing the t_{heavy} input in ADCNet.

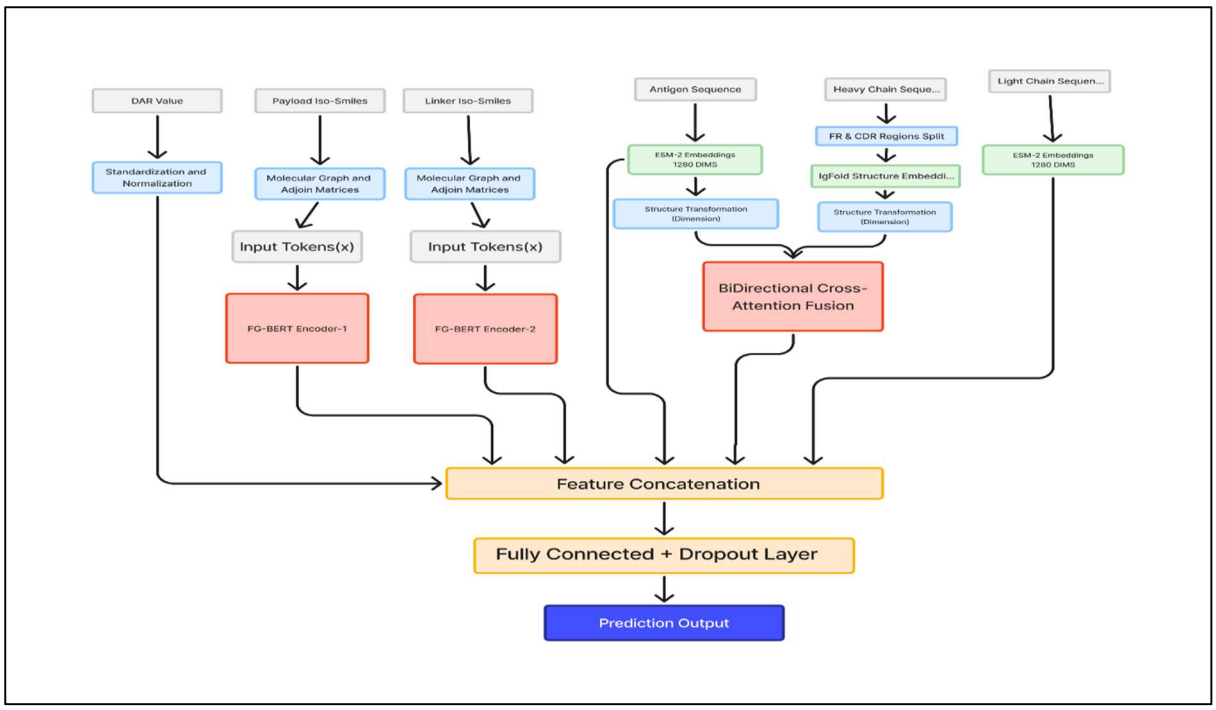
$$t_{AntiBinder} = \text{Antibinder}(\text{ Heavy Chain Sequence }, \text{ Antigen Sequence}), \quad (3).$$

The antigen feature vector $t_{antigen}$ and light chain feature vector t_{light} from ESM-2 ARE maintained in our design to retain more sequence-level information beyond the binding context. For the small molecule components (linker and payload), the feature vectors x_1^{AB} and x_2^{AB} respectively, are outputs from the modified FG-BERT model, which was ported to PyTorch. DAR value is scaled to the -1 ↔ 1 range to get t_{DAR}. The ABFormer feature concatenation is adapted as follows:

$$FF = concat\left(x_1^{AB}, \quad x_2^{AB}, \quad t_{AntiBinder}, \quad t_{light}, \quad t_{antigen}, \quad t_{DAR} \right), \quad (4)$$

Where $t_{AntiBinder}$ denotes the contextual binding representation of AntiBinder with 2592 dimensions, substituting for the individual heavy chain vector of the baseline ADCNet model, t_{light}, $t_{antigen}$, t_{DAR}, x_1^{AB} and x_2^{AB} denote light chain, antigen, DAR, linker and payload feature representations of dimensions 1280, 1280, 1, 256 and 256 respectively. The last feature fusion is done using a Multilayer Perceptron (MLP) [55] with input dimension of 5665, for predicting ADC activity, and formula as defined in Eq. (2).

Fig. 3. ABFormer's Structural Overview.



ABFormer's structural overview in Figure 3 represents a multimodal ADC prediction framework that integrates drug features (DAR, payload, and linker), antigen sequences, and antibody sequences (heavy/light chains). Features are encoded via FG-BERT and ESM-2 models, structurally transformed, and fused using BiDirectional Cross-Attention before passing through a fully connected network to generate the final prediction output.

2.3.2. ABFormer Training:

The selective training approach of the ABFormer model is such that only certain parts are subjected to parameter updates during training. To preserve the learned binding interaction representations, the pre-trained AntiBinder model, used for antibody-antigen binding data, remains unchanged during the training process. The parameters of the pre-trained ESM-2 and IgFold models [56, 57] are also left unchanged in their original state throughout the entire training process, as they are employed solely for deriving feature representations outside the training process. The training pipeline comprises the pre-trained FG-BERT model and the modified fully connected layers of the ADCNet architecture. The ABFormer architecture introduces approximately 300,000 additional trainable parameters compared to the baseline ADCNet model, primarily in the modified fully connected layers required to accommodate the expanded feature dimensionality from AntiBinder embeddings (2592-dim vs. 1280-dim for single heavy chain)To facilitate efficient batch-wise training with the PyTorch Distributed Data Parallel architecture [58], the adjacency matrix components were also stripped from the

original structure. AbFormer model was trained to a maximum of 100 epochs on the Adam optimiser [59] and initial learning rate of 1e-4 and a weight decay of 1e-4. These values were set through systematic empirical tuning with the objective to attain stable convergence and achieve the optimal generalization on the ADCdb set while also aligning with previous transformer-based molecular property prediction architecture. Training was carried out with a ReduceLROnPlateau schedule that monitored validation Area Under the Curve (AUC) and reduced learning rate by 0.5 times if improvement plateaued during 10 epochs. Moreover, an early stop approach with 30 epochs' patience was employed as a preventive against overfitting while validation performance was stable [60-62]. Along with these hyperparameters, trade-off of optimization stability between computational speed and accuracy was appropriately balanced so that AbFormer was able to deliver outcomes that were both predictable as well as reproducible.

2.4. Evaluation of model performance

The performance of the ABFormer model as well as that of the baseline ADCNet model is evaluated with a complete set of classification performance metrics that include positive predictive value (PPV), negative predictive value (NPV) [63], specificity (SP/TNR), sensitivity (SE/TPR/Recall), accuracy (ACC), F1 score, balanced accuracy (BA) [64], Matthews correlation coefficient (MCC) [65, 66], area under the precision-recall curve (PRAUC) [67], and area under the receiver operating characteristic curve (AUC-ROC) [68]. The formulae for the metrics are defined as follows:

$$PPV = \frac{TP}{TP+FP} \quad , \tag{5}$$

$$NPV = \frac{TN}{TN+FN} \quad , \tag{6}$$

$$SE = \frac{TP}{TP+FN} \quad , \tag{7}$$

$$SP = \frac{TN}{TN+FP} \quad , \tag{8}$$

$$ACC = \frac{TP+TN}{TP+TN+FP+FN} \quad , \tag{9}$$

$$F1 = \frac{2 \times TP}{2 \times TP + FP + FN} \quad , \tag{10}$$

$$BA = \frac{TPR+TNR}{2} = \frac{SE+SP}{2} \quad , \tag{11}$$

$$MCC = \frac{TP \times TN - FN \times FP}{\sqrt{(TP+FN) \times (TP+FP) \times (TN+FN) \times (TN+FP)}} \, . \tag{12}$$

Where TP, TN, FP, FN, TNR, and TPR represent true positive, true negative, false positive, false negative, actual negative rate, and true positive rate, respectively. Furthermore, AUC-

ROC is the area under the receiver operating characteristic curve plotted by calculating TPR vs. FPR at different discrimination thresholds, which measures the overall discriminative performance of the model. PRAUC represents the area under the precision-recall curve, which is particularly informative for evaluating performance on imbalanced datasets by plotting precision against recall at various threshold settings.

3. Results and discussion

3.1. ADC Modeling Dataset

Following the protocol developed in prior work, we trained our ADCNet part of the ABFormer framework using the ADCdb-derived dataset (the link to the dataset is given in the availability of data and materials section). The dataset includes 435 ADCs that vary in their structural and functional properties, with 154 distinct antibodies, 82 linkers, 71 payloads, 75 DAR values, and 66 antigens. Of these ADCs, 281 were assigned positive (active) labels, and 154 were assigned negative (inactive) labels. Structural variability within the components of the dataset is essential for model generalization [69, 70]

The 66 antigens fall into six distinct categories according to sequence alignment, with ERBB2 (106) [71], KIT (48) [72], FGFR2 (36) [73], TNFRSF1A (33) [74], and EGFR (22) [75] being the most common targets. Humanized IgG1-kappa antibodies are most prevalent within the repertoire of antibodies that indicate clinical preference. The chemical space analysis revealed that both linkers as well as payload molecules possess broad molecular weight ranges (linkers: 120.195–1702.11 Da; payload molecules: 233.097–1580.593 Da) as well as ranges of AlogP (linkers: -6.504–6.908; payload molecules: -4.675–7.406), thoroughly covering the chemical space.

Finally, the complete ADC modeling dataset was partitioned into training (80%), validation (10%), and test (10%) sets. The training set is utilized to develop a predictive model, while the test set serves to assess the model's final predicted performance.

3.2. Evaluation of Model Generalization with Rigorous Data Splitting

To thoroughly analyze the generalizability of our model fully and to ensure that our results were not artifacts of a specific data partition, we also carried out a scrupulous leave-pair-out validation analysis[76]. We carried out training and test processes for both benchmark ADCNet and ABFormer models on five different sets of 3 random seeds, utilizing different random initial seeds to test for stability and ability to generalize to unseen antibody-antigen pairs. Each of the 435 antibody-drug conjugates was assigned a distinct antibody-antigen pair identifier based on its heavy, light, and antigen, such that we identified 154 distinct pairs. Importantly, it was not the individual ADCs, but these pairs that were arbitrarily assigned to

the training, validation, and test splits under stratified sampling. There were 122 distinct pairs, which translates to 349 ADCs, in the training split, while both validation and test splits had 16 distinct pairs, with 44 and 42 ADCs, respectively. We confirmed that no overlapping of pairs between test sets and training sets occurred, ensuring a true estimation of generalizability to new candidates for therapy.

The results presented in Table 1 confirm that ABFormer exhibits stronger and more efficient generalization performance. ABFormer registered a remarkably high sensitivity (SE = 0.8933 compared with 0.7413), that is, it has a larger capability for accurate true positive ADC identification. This also comes with its high F1-score (0.8457 compared with 0.7920) [77], as well as Matthews Correlation Coefficient (MCC = 0.5952 compared to 0.5473) [65], which are principal measures of performance for class-imbalanced datasets. ABFormer, besides, presented a more accurate Negative Predictive Value (NPV = 0.8161 compared with 0.6851). Even though ADCNet was more precise (SP = 0.8117 compared to 0.6785) and Positive Predictive Value (PPV = 0.8565 compared to 0.8038), the vastly greater overall accuracy of ABFormer (ACC = 0.8064 compared to 0.7698) as well as Balanced Accuracy (BA = 0.7859 compared to 0.7765) [78] highlight its greater predictive capability. Lower standard deviations for ABFormer (e.g., SE: ±0.0377 compared to ±0.0600) represent more trustworthy performance. This experiment refutes concerns over data leakage and confirms that this enhancement observed in ABFormer derives from its architecture design.

Table 1. Comparison with ADCNet on Leaving Pair out Data Split.

Model	SE	SP	MCC	ACC	AUC	F1	BA	PRAUC	PPV	NPV
ABFormer	**0.8933 ± 0.0377**	**0.6785 ± 0.0363**	**0.5952 ± 0.0396**	**0.8064 ± 0.0192**	0.8162 ± 0.0071	**0.8457 ± 0.0179**	**0.7859 ± 0.0179**	**0.8667 ± 0.0062**	0.8038 ± 0.0148	**0.8161 ± 0.0458**
ADCNet	0.7413 ± 0.0600	0.8117 ± 0.0652	0.5473 ± 0.0267	0.7698 ± 0.0166	**0.8304 ± 0.0107**	0.7920 ± 0.0240	0.7765 ± 0.0143	0.8376 ± 0.0127	**0.8565 ± 0.0345**	0.6851 ± 0.0354

Leaving Pair out Data Split results were compared with ADCNet and are presented in Table 1 with best metrics highlighted in bold.

3.3. Model Performance Comparison

Our novel ABFormer model's performance was compared with that of the baseline deep learning model ADCNet as well as four popular machine learning (ML) models: Logistic Regression (LR) [79], Random Forest (RF), Support Vector Machine (SVM), and XGBoost

(XGB). The above ML models employ MACCS and Morgan fingerprints separately. All models are trained and validated on the same pre-curated ADC dataset with identical data splits, seeds, as well as initialization parameters to ensure a good comparison.

ABFormer achieved a robust balance of sensitivity (SE = 0.8851 ± 0.0430) and specificity (SP = 0.8000 ± 0.0544). This is a marked improvement over ADCNet, which itself had lower and more variable scores (SE = 0.7914 ± 0.1474, SP = 0.6261 ± 0.1866). The traditional machine learning models did overall poorly as well, particularly with respect to specificity, with some models such as RF-MORGAN as low as 0.5625 ± 0.1083. In overall accuracy as well, ABFormer had top scores (ACC = 0.8561 ± 0.0107), which was a marked improvement of 12.88% over ADCNet (ACC = 0.7273 ± 0.0371). Furthermore, it markedly improved on even the top-scoring machine learning model from this class represented here that was XGB-MACCS (ACC = 0.8258 ± 0.0473). In light of class imbalance in the dataset, we considered measures more appropriate for this situation. There, the strengths of ABFormer became most significant. It achieved the highest Matthews Correlation Coefficient (MCC = 0.6854 ± 0.0116), F1-score (0.8898 ± 0.0121), and Balanced Accuracy (BA = 0.8425 ± 0.0073). These performance levels well exceed ADCNet's results (MCC = 0.4404, F1 = 0.7827, BA = 0.7088) as well as top ML models on these corresponding measures. In addition, ABFormer had the highest PRAUC (0.9288 ± 0.0037), which reflects improved discriminative power under imbalanced conditions. Although the top models' AUC levels competed across models with XGB-MACCS exhibiting the largest figure (AUC = 0.8646 ± 0.0259), the overall strength of ABFormer across those imbalance-oriented measures reflects its overall robustness. Lastly, ABFormer exhibited the best-balanced and efficient predictive values (PPV = 0.8969 ± 0.0213, NPV = 0.7895 ± 0.0511). This represents a significant upgrade compared to ADCNet, which demonstrated low performance in terms of anticipating negatives (NPV = 0.6634 ± 0.1072). The traditional models performed even poorer with even lower reliability, especially in terms of NPV, with models such as LR-MACCS achieving lower than 0.60 values, thus signifying a critical deficiency in accurately predicting inactive compounds.

These findings verify that abformer achieves a substantial and stable gain over current methods for predicting ADC activity. Its remarkable performance on class imbalance sensitive measures can be explained from its design architecture that is capable of learning interactive context in ADC components as opposed to treating them as separate features. All traditional machine learning models as well as our model adcnet and abformer are trained with GPU [1× NVIDIA A100-SXM4-40GB] and CPU [AMD EPYC 7742 64-Core (128-thread) Processor].

Table 2: Performance evaluation results of ADCNet, traditional ML-based models, and ABFormer**.**

Model Name	SE	SP	ACC	AUC	PRAUC	F1	PPV	MCC	BA	NPV
LR - MACCS	0.6786±0.1988	0.7292±0.0361	0.6970±0.1389	0.7500±0.1879	0.8441±0.0976	0.7305±0.1498	0.8042±0.0720	0.4014±0.2329	0.7039±0.1164	0.5910±0.1636
LR-MORGAN	0.8095±0.1091	0.6458±0.2009	0.7500±0.0991	0.7909±0.0901	0.8639±0.0615	0.8043±0.0780	0.8063±0.1024	0.4660±0.2229	0.7277±0.1127	0.6727±0.1608
RF-MACCS	0.9048±0.0206	0.5625±0.1654	0.7803±0.0731	0.8199±0.0938	0.8853±0.0649	0.8409±0.0487	0.7865±0.0703	0.5055±0.1716	0.7336±0.0928	0.7615±0.0840
RF-MORGAN	**0.9286±0.0357**	0.5625±0.1083	0.7955±0.0601	0.8065±0.0749	0.8608±0.0427	0.8528±0.0420	0.7887±0.0478	0.5441±0.1412	0.7455±0.0702	0.8146±0.0955
SVM-MACCS	0.9167±0.0546	0.6458±0.0361	0.8182±0.0455	0.8155±0.0742	0.8575±0.0641	0.8648±0.0362	0.8188±0.0225	0.6000±0.1030	0.7813±0.0426	**0.8214±0.1017**
SVM-MORGAN	0.9167±0.0412	0.6250±0.1083	0.8106±0.0347	0.8095±0.0517	0.8603±0.0400	0.8605±0.0237	0.8125±0.0434	0.5830±0.0824	0.7708±0.0473	0.8167±0.0764
XGB-MACCS	0.8929±0.0357	0.7083±0.0955	0.8258±0.0473	**0.8646±0.0259**	0.9153±0.0177	0.8673±0.0348	0.8439±0.0466	0.6176±0.1073	0.8006±0.0562	0.7910±0.0685
XGB-MORGAN	0.8571±0.0945	0.6875±0.1083	0.7955±0.0909	0.8359±0.0370	0.8842±0.0068	0.8411±0.0745	0.8269±0.0620	0.5562±0.1910	0.7723±0.0922	0.7414±0.1413
ADCNet	0.7914±0.1474	0.6261±0.1866	0.7273±0.0371	0.8465±0.0608	0.9073±0.0454	0.7827±0.0537	0.8052±0.0737	0.4404±0.0300	0.7088±0.0283	0.6634±0.1072
ABFormer	0.8851±0.0430	**0.8000±0.0544**	**0.8561±0.0107**	0.8561±0.0107	**0.9288±0.0037**	**0.8898±0.0121**	**0.8969±0.0213**	**0.6854±0.0116**	**0.8425±0.0073**	0.7895±0.0511

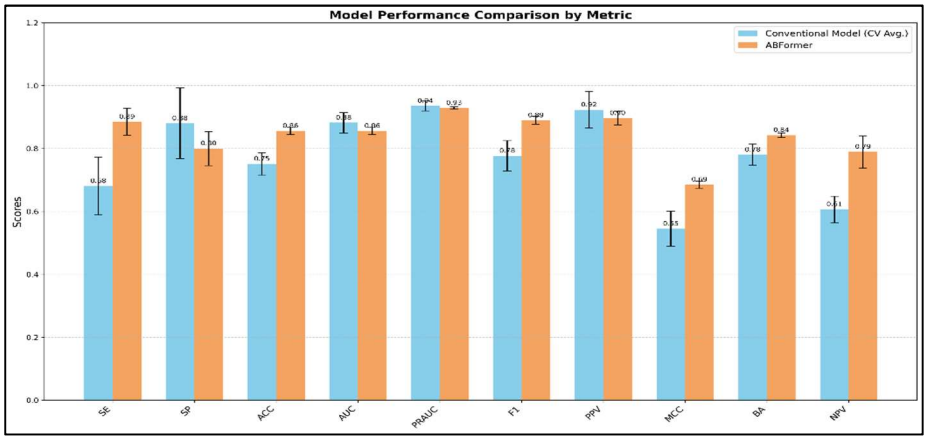
The performance results of all the models are presented in Table 2 and the best results are highlighted in bold.

3.4. Multi-level model validation

3.4.1. Cross-validation

In a bid to critically assess the generalizability of the ABFormer model and confirm that its performance was not reliant on arbitrary partitioning of the data, we adopted a 5-fold cross-validation (CV) protocol. This commonly used validation method is especially valuable for small-sized datasets since it provides a more stable as well as reliable estimation of the model's predictive capacity through its evaluation across different partitions of the data base. The outcome from cross-validation highlighted the robust and stable predictive ability of the model. The model performed outstandingly with characteristics of remarkable stability with average PRAUC of 0.9355 ± 0.0154 as well as an AUC of 0.8821 ± 0.0329. Furthermore, the model preserved outstanding overall performance with average Balanced Accuracy (BA) of 0.7805 ± 0.0335 as well as Accuracy (ACC) of 0.7511 ± 0.0356. Cross-validation indicated fair variance within class-specific measures particularly Sensitivity (SE 0.6810 ± 0.0908) as well as Specificity (SP 0.8800 ± 0.1127). Conversely, top measures from the level of aggregative analysis presented a high level of stability. Such overall stability reported even with the expected variability that is usually seen with smaller datasets shows extreme robustness of the ABFormer model as well as its reliable ability to generalize.

Fig. 4: Cross-validation results comparison.



The bar chart in Figure 4 illustrates the relative performance of the ABformer model on the test set and through cross-validation across different performance metrics. The performance metrics—Sensitivity (SE), Specificity (SP), Matthews Correlation Coefficient (MCC), Accuracy (ACC), Area Under the Curve (AUC), F1 Score, Balanced Accuracy (BA), Precision-Recall AUC (PRAUC), Positive Predictive Value (PPV), Negative Predictive Value (NPV), respectively—are shown through adjacent bars with green highlighting test set performance as well as orange highlighting cross-validation results.

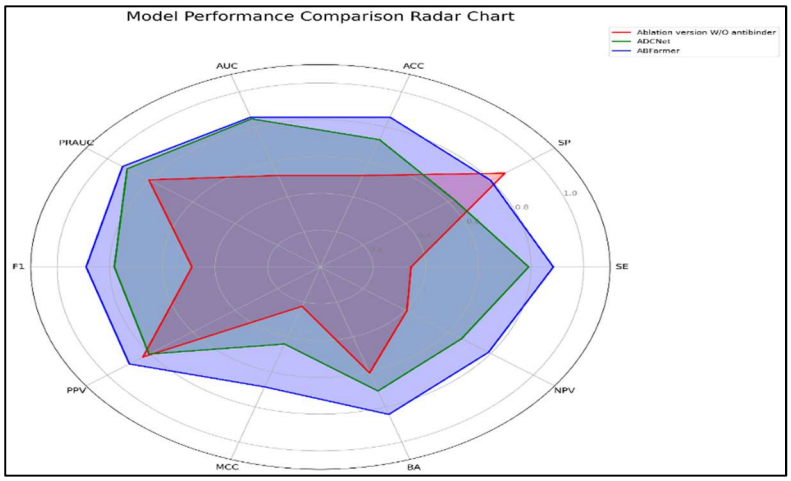
3.4.2. Model architecture validation based on ablation experiments

To isolate and measure the unique contribution of our chief architectural innovation, an ablation study was conducted. An ablation variant was created by omitting the contextual antibody-antigen binding depiction, otherwise known as the AntiBinder module, from the feature concatenation step. This variant was left to predict on its own, along with the remaining ADC features, including the linker, payload, antibody light chain, antigen, and DAR embeddings. The ablated model was trained and tested according to the complete ABFormer protocol to allow a direct and fair comparison.

The experimental results presented in Fig. 5 exhibit notable performance degeneracy after ablation of the AntiBinder module. The overall ABFormer model was capable of obtaining a Matthews correlation coefficient (MCC) score of 0.6854 ± 0.0116 and a balanced accuracy (BA) score of 0.8425 ± 0.0073, reflecting its outstanding discriminative ability. The ablated model's performance was also subpar, with its MCC value falling to 0.2251 ± 0.0000 and its BA value falling to 0.6057 ± 0.0000, with resultant relative performance losses of roughly 67% and 28%, respectively. The performance is lower than that of the baseline ADCNet model (MCC = 0.4404, BA = 0.7088), thus highlighting the essential role that is played by the AntiBinder module in allowing ABFormer to perform outstandingly.

This decreased performance was seen consistently with all other criteria for evaluation. The AUC-ROC (0.5227) as well as accuracy (0.5227) of the ablated variant represented a predictive power that was slightly better than chance alone, significantly in contrast with both significantly high scores at both criteria (0.8561 both) from the complete ABFormer model. In the same manner, the F1-score reduced from 0.8898 for ABFormer down to just 0.4878 in the ablated form. These findings conclusively illustrate that contextual binding information represented in the AntiBinder module is essential to attaining high predictive accuracy. In combination, these findings verify that the resultant antibody-antigen binding element integrated into the structure of ABFormer is not just helpful but, in fact, indispensable. Its removal significantly handicaps the predictive capability of the model, thereby confirming the entire structure of ABFormer as a stable and efficient scaffold adopted in ADC activity prediction.

Fig. 5: Performance Visualisation of ABFormer and Ablation Variant.



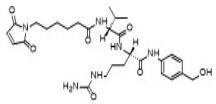
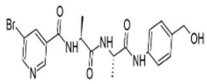
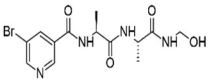
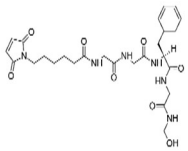
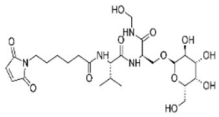
The radar chart in the Figure 5 compares the performance of the original ABFormer model with an ablation variant and ADCNet across multiple evaluation metrics: Sensitivity (SE), Specificity (SP), Matthews Correlation Coefficient (MCC), Accuracy (ACC), Area Under the Curve (AUC), F1 Score, Balanced Accuracy (BA), Precision-Recall AUC (PRAUC), Positive Predictive Value (PPV), and Negative Predictive Value (NPV).

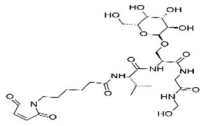
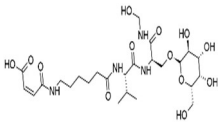
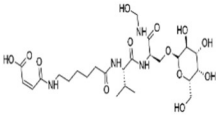
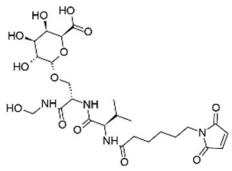
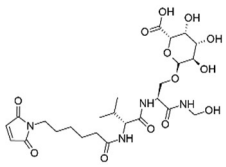
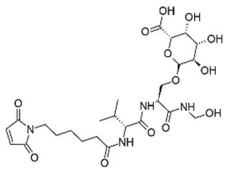
3.5. Testing with an Independent External Data Set

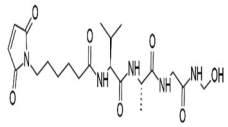
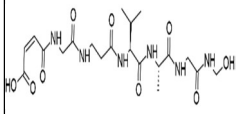
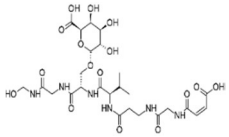
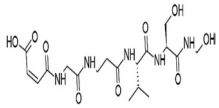
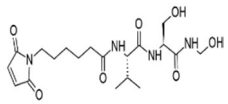
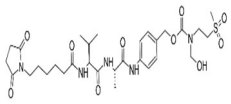
To extensively verify the in-the-wild generalization capability of the ABFormer and the baseline ADCNet models, we also conducted an activity predicability evaluation with an independent and external set. The original set, generated from recent literature and a patent by the ADCNet authors, consists of 21 new ADCs that were not present in the original training and validation sets. Systematically, the novelty of the external set was confirmed with similarity analysis, which indicated moderate-to-high dissimilarity for these 21 ADCs in the modeling-data space. Thus, the set represents a rigorous and objective test set for evaluating the models' predictive activity for unseen ADC constructs.

The complete predictive outcomes concerning the external dataset are summarized in Table 3. The baseline ADCNet model accurately predicted the anticancer activity of 19 among the 21 ADCs, achieving a remarkable prediction accuracy of 90.48%. In an impressive leap, our ABFormer model demonstrated its best performance by achieving a perfect accuracy of 100%, correctly predicting the activity of all the ADCs in a set of 21. These predictions not only confirm the good generalizability of our model in forecasting the activity of various antibody-drug conjugates (ADCs) but also point to its stronger predictive capacity under realistic discovery circumstances than existing state-of-the-art models.

558 **Table 3.** Prediction Probabilities on the Independent external Dataset.

No.	Antibody	Antigen	Payload	Linker	DAR	Similarity score	Bioactivity value	Label	ADCNet	AbFormer
1	H1D8	CD44v5	MMAE		3.7	0.7867	IC_{50} = 8.24 µg/mL	1	0.972	0.9915
2	Cetuximab	EGFR	Gemcitabine		6	0.4047	IC_{50} = 10 nM	1	0.9956	0.6598
3	Trastuzumab	HER2	Gemcitabine		6	0.4051	IC_{50} = 150 nM	0	0.9907	0.3043
4	Sacituzumab	TROP2	Auristatin E		6.1	0.724	IC_{50} = 0.03–0.09 nM	1	0.9839	0.8564
5	Sacituzumab	TROP2	DM1		6.2	0.6541	IC_{50} = 0.05–0.1 nM	1	0.9991	0.821
6	Sacituzumab	TROP2	DXd		7.3	0.5757	IC_{50} = 0.4–0.7 nM	1	0.9817	0.8558
7	Ifinatamab	CD276	DXd		8.12	0.9662	EC_{50} = 0.022 – 3.764 nM	1	0.9889	0.9934
8	Ifinatamab	CD276	DXd		8.76	0.8152	EC_{50} = 0.008 – 3.291 nM	1	0.9326	0.9935

9	Ifinatamab	CD276	DXd		7.76	0.8019	EC_{50} = 0.018 – 2.291 nM	1	0.9648	0.9935
10	Ifinatamab	CD276	DXd		7.98	0.7733	EC_{50} = 0.038 – 1.245 nM	1	0.9946	0.9934
11	Ifinatamab	CD276	DXd		8.12	0.7616	EC_{50} = 0.013 –0.53 nM	1	0.995	0.9935
12	Ifinatamab	CD276	DXd		8.06	0.8107	EC_{50} = 0.007 – 0.894 nM	1	0.9847	0.9934
13	Ifinatamab	CD276	DXd		8.02	0.7271	EC_{50} = 0.015 – 1.042 nM	1	0.9946	0.9934
14	Ifinatamab	CD276	DXd		8.18	0.7976	EC_{50} = 0.024 – 1.647 nM	1	0.991	0.9935

15	Ifinatamab	CD276	DXd		8.52	0.9125	EC_{50} = 0.025 – 1.073 nM	1	0.9855	0.9934
16	Ifinatamab	CD276	DXd		8.08	0.8544	EC_{50} = 0.025 – 1.234 nM	1	0.9958	0.9934
17	Ifinatamab	CD276	DXd		7.76	0.7273	EC_{50} = 0.029 – 1.094 nM	1	0.9958	0.9935
18	Ifinatamab	CD276	DXd		7.78	0.8074	EC_{50} = 0.01– 0.861 nM	1	0.9958	0.9934
19	Ifinatamab	CD276	DXd		8.56	0.8914	EC_{50} = 0.033 – 0.511 nM	1	0.9616	0.9934
20	hLL2-C220	CD22	SG3249		1.52	0.8196	IC_{50} = 0.02– 51.7 nM	1	0.9998	0.9989
21	FZ-A038	DLL3	DXd		8	0.6023	IC_{50} = 0.15– 4.29 nM	1	0.0196	0.9994

The probability scores of ABFormer and ADCNet on the independent benchmark dataset are displayed in the Table 3.

4. Future Improvements

Although ABFormer demonstrates significant performance gains over existing approaches, several avenues for further improvement remain open. One possible future consideration of model optimization is taking advantage of parameter-efficient fine-tuning techniques, particularly Quantized Low-Rank Adaptation(QLoRA) The utilization of QLoRA would allow us to perform efficient fine-tuning of large pre-trained models (ESM-2, FG-BERT) within our system while still maintaining memory efficiency as well as minimizing the risk of overfitting on the limited ADC dataset [80, 81].

Furthermore, conducting comprehensive fine-tuning of the AntiBinder component, specifically targeting ADC-related antibody-antigen pairs, has the potential to significantly improve the accuracy of binding predictions. With the increasing availability of ADC datasets, implementing end-to-end fine-tuning of the entire ABFormer architecture may yield further enhancements by enabling all components to more effectively adapt to the ADC prediction task.

Integrating more structural information, such as linker conformational flexibility as well as payload-target interaction profiles, is another potential research direction. Further, development of multi-task learning approaches that concurrently forecast antibody-drug conjugate (ADC) activity, toxicity, and pharmacokinetic properties may offer richer ADC design advice.

5. Conclusion

The forecasting of Antibody-Drug Conjugate (ADC) activity is hindered by complicated, synergistic relationships that are often modeled as independent component features. This work demonstrates that forecasting of ADC activity is substantially enhanced by explicit antibody-antigen binding interface modeling.

The ABFormer model, as introduced, integrates context antibody-antigen binding characteristics of an AntiBinder with point-wise encoding of the linker, payload, as well as other ADC components. The ABFormer model includes the mutually interacting molecule dynamics that are significant for ADC effectiveness, which was lacking in prior models.

ABFormer exhibits considerably enhanced yet stable behavior over the baseline control ADCNet, with a balanced accuracy of 84.25% as well as a PRAUC of 92.88%. Its generalizability holds also with harsh leave-pair-out validation, as well as a perfect 100% accuracy rate on a different external test set. Of most interest, ablation experiments essentially confirm the role of context binding module itself; its removal causes a dramatic loss of predictive capability with MCC dropping to 0.2251 from 0.6854.

These findings establish that it is imperative to model component interdependency, especially at binding interface, to obtain precise in silico antibody-drug conjugate (ADC) screening. ABFormer provides a benchmarked computational framework that aids in rapid design of efficient ADC therapeutics. Future studies will focus on generalizing this framework to accommodate multi-task toxicity and pharmacokinetic profile prediction under concurrent conditions.

List of abbreviations

ACC - Accuracy

ADC - Antibody Drug Conjugates

ADCdb - Antibody Drug Conjugates databank

AUC - Area Under Curve

BA - Balanced Accuracy

DAR - Drug-to-Antibody Ratio

ESM2 - Evolutionary Scale Modeling version 2

Fc - Fragment Crystallisable

FFN - Feed Forward Network

FG-BERT – Fine Grained - Bidirectional Encoder Representations from Transformers

FN - False Negative

FP - False Positive

HER2 – Human Epidermal Growth Factor Receptor 2

LR - Logistic Regression

MCC – Matthews Correlation Coefficient

MET dataset – Mesenchymal-Epithelial transition gene dataset

MHA - Multi-Head Attention

ML - Machine Learning

MLP – Multi Layer Perceptron

NPV – Negative Predictive Value

PPV – Positive Predictive Value

PRAUC – Precision-Recall AUC

QLoRA – Quantized Low-Rank Adaptation

RF - Random Forest

SE - Sensitivity

SP - Specificity

SVM - Support Vector Machine

TAA – Tumor Associated Antigen

TN - True Negatives

TNR - True Negative Rate

TP - True Positives

TPR - True Positive Rate

XGB - XG Boost

Declarations

Ethics approval and consent to participate

Not applicable

Consent for publication

Not applicable

Availability of Data and Materials

The database used in this study was obtained from the IDRBLab Antibody–Drug Conjugate Database (ADCDB) available at https://adcdb.idrblab.net/ .

Code Availability

The data and code are available at https://github.com/drugparadigm/ABFormer

Competing interests

The authors declare that they have no competing interests.

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Author Contributions

RK: Investigation; validation; methodology; visualization; writing—original draft; formal analysis. **VL:** Data curation; formal analysis; methodology; validation; visualization. **SG:** Conceptualization; methodology; validation; visualization; formal analysis; data curation; software; supervision; formal analysis; writing—review and editing. **VK:** Conceptualization; methodology; writing—review and editing; supervision; formal analysis.

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REFERENCES

1. Fu Z, Li S, Han S, Shi C, Zhang Y. Antibody drug conjugate: the "biological missile" for targeted cancer therapy. Signal Transduction and Targeted Therapy. 2022 2022/03/22;7(1):93.
2. Tang H, Liu Y, Yu Z, Sun M, Lin L, Liu W, et al. The Analysis of Key Factors Related to ADCs Structural Design. Frontiers in Pharmacology. [Review]. 2019 2019-April-24;Volume 10 - 2019.
3. Zahavi D, Weiner L. Monoclonal Antibodies in Cancer Therapy. Antibodies. 2020;9(3):34.
4. Lei Y, Zheng M, Chen P, Seng Ng C, Peng Loh T, Liu H. Linker Design for the Antibody Drug Conjugates: A Comprehensive Review. ChemMedChem. 2025;20(15):e202500262.
5. Sun H, Wienkers LC, Lee A. Beyond cytotoxic potency: disposition features required to design ADC payload. Xenobiotica. 2024 2024/09/13;54(8):442-57.
6. Izzo D, Ascione L, Guidi L, Marsicano RM, Koukoutzeli C, Trapani D, et al. Innovative payloads for ADCs in cancer treatment: moving beyond the selective delivery of chemotherapy. Therapeutic Advances in Medical Oncology. 2025;17:17588359241309461.
7. Valsasina B, Orsini P, Terenghi C, Ocana A. Present Scenario and Future Landscape of Payloads for ADCs: Focus on DNA-Interacting Agents. Pharmaceuticals. 2024;17(10):1338.
8. Dong W, Wang W, Cao C. The Evolution of Antibody-Drug Conjugates: Toward Accurate DAR and Multi-specificity. ChemMedChem. [Review]. 2024 Sep 2;19(17):e202400109.
9. Pettinato MC. Introduction to Antibody-Drug Conjugates. Antibodies. [Review]. 2021 Oct 27;10(4).
10. Lv Y, Cui X, Li T, Liu C, Wang A, Wang T, et al. Mechanism of action and future perspectives of ADCs in combination with immune checkpoint inhibitors for solid tumors. Clin Exp Med. [Review]. 2025 May 4;25(1):139.
11. Maiti R, Patel B, Patel N, Patel M, Patel A, Dhanesha N. Antibody drug conjugates as targeted cancer therapy: past development, present challenges and future opportunities. Arch Pharm Res. [Review]. 2023 May;46(5):361-88.
12. Luo S, Su Y, Peng X, Wang S, Peng J, Ma J. Antigen-specific antibody design and optimization with diffusion-based generative models for protein structures. Proceedings of the 36th International Conference on Neural Information Processing Systems; New Orleans, LA, USA: Curran Associates Inc.; 2022. p. Article 709.
13. Martinkus K, Ludwiczak J, Cho K, Liang W-C, Lafrance-Vanasse J, Hotzel I, et al. AbDiffuser: full-atom generation of in vitro functioning antibodies. Proceedings of the 37th International Conference on Neural Information Processing Systems; New Orleans, LA, USA: Curran Associates Inc.; 2023. p. Article 1774.
14. He H, He B, Guan L, Zhao Y, Jiang F, Chen G, et al. De novo generation of SARS-CoV-2 antibody CDRH3 with a pre-trained generative large language model. Nature Communications. 2024 2024/08/10;15(1):6867.
15. Hie BL, Shanker VR, Xu D, Bruun TUJ, Weidenbacher PA, Tang S, et al. Efficient evolution of human antibodies from general protein language models. Nature Biotechnology. 2024 2024/02/01;42(2):275-83.
16. Li M, Shi Y, Hu S, Guo P, Wan W, Zhang LY, et al. MVSF-AB: accurate antibody-antigen binding affinity prediction via multi-view sequence feature learning. Bioinformatics. 2025 May 6;41(5).
17. He H, Chen G, Chen CY-C. Integrating sequence and graph information for enhanced drug-target affinity prediction. Science China Information Sciences. 2024 2024/01/25;67(2):129101.
18. He H, Chen G, Tang Z, Chen CY-C. Dual modality feature fused neural network integrating binding site information for drug target affinity prediction. npj Digital Medicine. 2025 2025/01/28;8(1):67.
19. Mukit H, Hossain S, Farabi MM, Chowdhury MZ, Pritom AI, Rana HK. SMILES2DTA: a CNN-based approach for identifying drug candidates and predicting drug-target binding affinity. Neural Comput Appl. 2024;37:2891-910.
20. McCombs JR, Owen SC. Antibody drug conjugates: design and selection of linker, payload and conjugation chemistry. Aaps J. [Review]. 2015 Mar;17(2):339-51.
21. Chen L, Li B, Chen Y, Lin M, Zhang S, Li C, et al. ADCNet: a unified framework for predicting the activity of antibody-drug conjugates. Brief Bioinform. 2025 May 3;26(3).
22. Pysz I, Jackson PJM, Thurston DE. Introduction to Antibody–Drug Conjugates (ADCs). In: Thurston DE, Jackson PJM, editors. Cytotoxic Payloads for Antibody – Drug Conjugates: The Royal Society of Chemistry; 2019. p. 0.

23. Li W, Zhao Y, Chen X, Xiao Y, Qin Y. Detecting Alzheimer's Disease on Small Dataset: A Knowledge Transfer Perspective. IEEE J Biomed Health Inform. [Research Support, N.I.H., Extramural Research Support, Non-U.S. Gov't
Research Support, U.S. Gov't, Non-P.H.S.]. 2019 May;23(3):1234-42.
24. Shen L, Sun X, Chen Z, Guo Y, Shen Z, Song Y, et al. ADCdb: the database of antibody-drug conjugates. Nucleic Acids Res. 2024 Jan 5;52(D1):D1097-D109.
25. Li F, Hu Q, Zhang X, Sun R, Liu Z, Wu S, et al. DeepPROTACs is a deep learning-based targeted degradation predictor for PROTACs. Nat Commun. [Research Support, Non-U.S. Gov't]. 2022 Nov 21;13(1):7133.
26. Zhao L, He Q, Song H, Zhou T, Luo A, Wen Z, et al. Protein A-like Peptide Design Based on Diffusion and ESM2 Models. Molecules. 2024 Oct 21;29(20).
27. Lin Z, Akin H, Rao R, Hie B, Zhu Z, Lu W, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. Science. 2023 Mar 17;379(6637):1123-30.
28. Agarap AF. Deep Learning using Rectified Linear Units (ReLU). ArXiv. 2018;abs/1803.08375.
29. Pratiwi H, Windarto A, Susliansyah S, Aria R, Susilowati S, Rahayu L, et al. Sigmoid Activation Function in Selecting the Best Model of Artificial Neural Networks. Journal of Physics: Conference Series. 2020;1471:012010.
30. Li B, Lin M, Chen T, Wang L. FG-BERT: a generalized and self-supervised functional group-based molecular representation learning framework for properties prediction. Brief Bioinform. [Research Support, Non-U.S. Gov't]. 2023 Sep 22;24(6).
31. Wei J, Zhang X, Ji Z, Li J, Wei Z. Deploying and scaling distributed parallel deep neural networks on the Tianhe-3 prototype system. Sci Rep. [Research Support, Non-U.S. Gov't]. 2021 Oct 12;11(1):20244.
32. Sibilano E, Buongiorno D, Lassi M, Grippo A, Bessi V, Sorbi S, et al. Understanding the Role of Self-Attention in a Transformer Model for the Discrimination of SCD From MCI Using Resting-State EEG. IEEE J Biomed Health Inform. [Research Support, Non-U.S. Gov't]. 2024 Jun;28(6):3422-33.
33. Choi SR, Lee M. Transformer Architecture and Attention Mechanisms in Genome Data Analysis: A Comprehensive Review. Biology (Basel). [Review]. 2023 Jul 22;12(7).
34. Zhang K, Tao Y, Wang F. AntiBinder: utilizing bidirectional attention and hybrid encoding for precise antibody-antigen interaction prediction. Brief Bioinform. 2024 Nov 22;26(1).
35. Huang Y, Zhang Z, Zhou Y. AbAgIntPre: A deep learning method for predicting antibody-antigen interactions based on sequence information. Front Immunol. [Research Support, Non-U.S. Gov't]. 2022;13:1053617.
36. Ruffolo JA, Chu LS, Mahajan SP, Gray JJ. Fast, accurate antibody structure prediction from deep learning on massive set of natural antibodies. Nat Commun. [Research Support, N.I.H., Extramural Research Support, Non-U.S. Gov't]. 2023 Apr 25;14(1):2389.
37. Makowski EK, Kinnunen PC, Huang J, Wu L, Smith MD, Wang T, et al. Co-optimization of therapeutic antibody affinity and specificity using machine learning models that generalize to novel mutational space. Nat Commun. [Research Support, N.I.H., Extramural
Research Support, Non-U.S. Gov't
Research Support, U.S. Gov't, Non-P.H.S.]. 2022 Jul 1;13(1):3788.
38. Qian CN, Mei Y, Zhang J. Cancer metastasis: issues and challenges. Chin J Cancer. [Editorial]. 2017 Apr 3;36(1):38.
39. Trail PA, Dubowchik GM, Lowinger TB. Antibody drug conjugates for treatment of breast cancer: Novel targets and diverse approaches in ADC design. Pharmacol Ther. [Review]. 2018 Jan;181:126-42.
40. Zhu Z, Ashrafian H, Tabrizi NM, Matas E, Girard L, Ma H, et al. Antibody numbering schemes: advances, comparisons and tools for antibody engineering. Protein Eng Des Sel. [Review]. 2025 Jan 10;38.
41. Dunbar J, Deane CM. ANARCI: antigen receptor numbering and receptor classification. Bioinformatics. [Research Support, Non-U.S. Gov't]. 2016 Jan 15;32(2):298-300.
42. Kumar M, Patel AK, Biswas M, Shitharth S. Attention-based bidirectional-long short-term memory for abnormal human activity detection. Sci Rep. 2023 Sep 2;13(1):14442.

43. Li Y, Fan Z, Rao J, Chen Z, Chu Q, Zheng M, et al. An overview of recent advances and challenges in predicting compound-protein interaction (CPI). Med Rev (2021). [Review]. 2023 Dec;3(6):465-86.
44. Shi SR, Liu C, Perez J, Taylor CR. Protein-embedding technique: a potential approach to standardization of immunohistochemistry for formalin-fixed, paraffin-embedded tissue sections. J Histochem Cytochem. [Research Support, N.I.H., Extramural Research Support, U.S. Gov't, P.H.S.]. 2005 Sep;53(9):1167-70.
45. Pishesha N, Harmand TJ, Ploegh HL. A guide to antigen processing and presentation. Nat Rev Immunol. [Review]. 2022 Dec;22(12):751-64.
46. Xie X, Chen L, Qin S, Zha F, Fan X. Bidirectional feature pyramid attention-based temporal convolutional network model for motor imagery electroencephalogram classification. Front Neurorobot. 2024;18:1343249.
47. Gu M, Yang W, Liu M. Prediction of antibody-antigen interaction based on backbone aware with invariant point attention. BMC Bioinformatics. 2024 Nov 6;25(1):348.
48. Jin R, Ye Q, Wang J, Cao Z, Jiang D, Wang T, et al. AttABseq: an attention-based deep learning prediction method for antigen-antibody binding affinity changes based on protein sequences. Brief Bioinform. 2024 May 23;25(4).
49. Wu Z, Zhang X, Li F, Wang S, Li J. TransRender: a transformer-based boundary rendering segmentation network for stroke lesions. Front Neurosci. 2023;17:1259677.
50. Zhang Y, Liu C, Liu M, Liu T, Lin H, Huang CB, et al. Attention is all you need: utilizing attention in AI-enabled drug discovery. Brief Bioinform. [Research Support, Non-U.S. Gov't Review]. 2023 Nov 22;25(1).
51. Yan J, Wang X, Cai J, Qin Q, Yang H, Wang Q, et al. Medical image segmentation model based on triple gate MultiLayer perceptron. Sci Rep. [Research Support, Non-U.S. Gov't]. 2022 Apr 12;12(1):6103.
52. Ozyoruk KB, Harmon SA, Lay NS, Yilmaz EC, Bagci U, Citrin DE, et al. AI-ADC: Channel and Spatial Attention-Based Contrastive Learning to Generate ADC Maps from T2W MRI for Prostate Cancer Detection. J Pers Med. 2024 Oct 9;14(10).
53. Gao M, Skolnick J. Improved deep learning prediction of antigen-antibody interactions. Proc Natl Acad Sci U S A. 2024 Oct 8;121(41):e2410529121.
54. Xia Y, Wang Z, Huang F, Xiong Z, Wang Y, Qiu M, et al. DeepInterAware: Deep Interaction Interface-Aware Network for Improving Antigen-Antibody Interaction Prediction from Sequence Data. Adv Sci (Weinh). 2025 Apr;12(13):e2412533.
55. Murtagh F. Multilayer perceptrons for classification and regression. Neurocomputing. 1991 1991/07/01/;2(5):183-97.
56. Jing H, Gao Z, Xu S, Shen T, Peng Z, He S, et al. Accurate prediction of antibody function and structure using bio-inspired antibody language model. Brief Bioinform. 2024 May 23;25(4).
57. Meng F, Zhou N, Hu G, Liu R, Zhang Y, Jing M, et al. A comprehensive overview of recent advances in generative models for antibodies. Comput Struct Biotechnol J. [Review]. 2024 Dec;23:2648-60.
58. Tahir H, Jung ES. Comparative Study on Distributed Lightweight Deep Learning Models for Road Pothole Detection. Sensors (Basel). 2023 Apr 27;23(9).
59. Jais IKM, Ismail AR, Nisa SQ. Adam Optimization Algorithm for Wide and Deep Neural Network. Knowl Eng Data Sci. 2019;2:41-6.
60. Tucat M, Mukherjee A. Regularized Gradient Clipping Provably Trains Wide and Deep Neural Networks. Trans Mach Learn Res. 2024;2025.
61. Paguada S, Batina L, Buhan I, Armendariz I. Being Patient and Persistent: Optimizing An Early Stopping Strategy for Deep Learning in Profiled Attacks. IEEE Trans Comput. 2025;74(3):875–86.
62. Ying X. An Overview of Overfitting and its Solutions. Journal of Physics: Conference Series. 2019;1168:022022.
63. Monaghan TF, Rahman SN, Agudelo CW, Wein AJ, Lazar JM, Everaert K, et al. Foundational Statistical Principles in Medical Research: Sensitivity, Specificity, Positive Predictive Value, and Negative Predictive Value. Medicina (Kaunas). [Review]. 2021 May 16;57(5).

64. Varoquaux G, Colliot O. Evaluating Machine Learning Models and Their Diagnostic Value. In: Colliot O, editor. Machine Learning for Brain Disorders. New York, NY2023. p. 601-30.
65. Chicco D, Jurman G. The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. BMC Genomics. 2020 Jan 2;21(1):6.
66. Chicco D, Totsch N, Jurman G. The Matthews correlation coefficient (MCC) is more reliable than balanced accuracy, bookmaker informedness, and markedness in two-class confusion matrix evaluation. BioData Min. 2021 Feb 4;14(1):13.
67. Martin B, Bennett TD, DeWitt PE, Russell S, Sanchez-Pinto LN. Use of the Area Under the Precision-Recall Curve to Evaluate Prediction Models of Rare Critical Illness Events. Pediatr Crit Care Med. 2025 Jun 1;26(6):e855-e9.
68. Habibzadeh F. Diagnostic tests performance indices: an overview. Biochem Med (Zagreb). [Review]. 2025 Feb 15;35(1):010101.
69. Raghavan P, Haas BC, Ruos ME, Schleinitz J, Doyle AG, Reisman SE, et al. Dataset Design for Building Models of Chemical Reactivity. ACS Cent Sci. [Review]. 2023 Dec 27;9(12):2196-204.
70. Aliferis C, Simon G. Overfitting, Underfitting and General Model Overconfidence and Under-Performance Pitfalls and Best Practices in Machine Learning and AI. In: Simon GJ, Aliferis C, editors. Artificial Intelligence and Machine Learning in Health Care and Medical Sciences: Best Practices and Pitfalls. Cham (CH)2024. p. 477-524.
71. Raghav KPS, Moasser MM. Molecular Pathways and Mechanisms of HER2 in Cancer Therapy. Clin Cancer Res. [Research Support, N.I.H., Extramural
Review]. 2023 Jul 5;29(13):2351-61.
72. Zhou S, Abdihamid O, Tan F, Zhou H, Liu H, Li Z, et al. KIT mutations and expression: current knowledge and new insights for overcoming IM resistance in GIST. Cell Commun Signal. [Research Support, Non-U.S. Gov't
Review
Video-Audio Media]. 2024 Feb 27;22(1):153.
73. Dixit G, Gonzalez-Bosquet J, Skurski J, Devor EJ, Dickerson EB, Nothnick WB, et al. FGFR2 mutations promote endometrial cancer progression through dual engagement of EGFR and Notch signalling pathways. Clin Transl Med. [Research Support, N.I.H., Extramural
Research Support, Non-U.S. Gov't
Research Support, U.S. Gov't, Non-P.H.S.]. 2023 May;13(5):e1223.
74. Assrawi E, Louvrier C, El Khouri E, Delaleu J, Copin B, Dastot-Le Moal F, et al. Mosaic variants in TNFRSF1A: an emerging cause of tumour necrosis factor receptor-associated periodic syndrome. Rheumatology (Oxford). [Case Reports
Research Support, Non-U.S. Gov't]. 2022 Dec 23;62(1):473-9.
75. Uribe ML, Marrocco I, Yarden Y. EGFR in Cancer: Signaling Mechanisms, Drugs, and Acquired Resistance. Cancers (Basel). [Review]. 2021 Jun 1;13(11).
76. Geroldinger A, Lusa L, Nold M, Heinze G. Leave-one-out cross-validation, penalization, and differential bias of some prediction model performance measures-a simulation study. Diagn Progn Res. 2023 May 2;7(1):9.
77. Yacouby R, Axman D, editors. Probabilistic Extension of Precision, Recall, and F1 Score for More Thorough Evaluation of Classification Models. EVAL4NLP; 2020.
78. García V, Mollineda RA, Sánchez JS, editors. Index of Balanced Accuracy: A Performance Measure for Skewed Class Distributions2009; Berlin, Heidelberg: Springer Berlin Heidelberg.
79. Zabor EC, Reddy CA, Tendulkar RD, Patil S. Logistic Regression in Clinical Studies. Int J Radiat Oncol Biol Phys. 2022 Feb 1;112(2):271-7.
80. Sledzieski S, Kshirsagar M, Baek M, Dodhia R, Lavista Ferres J, Berger B. Democratizing protein language models with parameter-efficient fine-tuning. Proc Natl Acad Sci U S A. 2024 Jun 25;121(26):e2405840121.
81. Dietrich J, Hollstein A. Performance and Reproducibility of Large Language Models in Named Entity Recognition: Considerations for the Use in Controlled Environments. Drug Safety. 2025 2025/03/01;48(3):287-303.

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